Single-Cell with Seurat

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Introduction

Seurat is the most widely used R framework for single-cell RNA-seq (scRNA-seq) analysis. A standard Seurat workflow loads a count matrix from Cell Ranger or a similar source, performs quality control on per-cell metrics, normalises expression, selects variable features, scales, runs dimensionality reduction (PCA + UMAP/t-SNE), clusters cells, and identifies marker genes. The framework's strength is its end-to-end coverage and large user community; the alternative Bioconductor stack (`SingleCellExperiment` + `scran`) offers comparable functionality with tighter integration into the broader Bioconductor ecosystem.

Prerequisites

A working understanding of single-cell RNA-seq experimental design, count matrices from droplet-based protocols (10x Chromium), and basic clustering and dimensionality-reduction concepts.

Theory

Seurat stores cells $\times$ genes counts and derived matrices in a `Seurat` object with structured slots for raw counts, normalised data, variable features, scaled data, dimensional reductions, and per-cell metadata. The standard pipeline:

1. **QC**: filter cells by minimum gene count, maximum percent mitochondrial reads, library size thresholds.
2. **Normalisation**: log-normalisation (`NormalizeData`) or regularised regression (`SCTransform`).
3. **Variable-feature selection**: identify genes with biologically interesting variability.
4. **Scaling**: centre and unit-variance scale variable features.
5. **PCA**: linear dimensionality reduction on scaled variable features.
6. **Graph-based clustering**: shared-nearest-neighbour graph + Louvain or Leiden community detection.
7. **UMAP / t-SNE**: non-linear visualisation only — not used for clustering decisions.

Assumptions

Each droplet is mostly single-cell (low doublet rate, typically below 5 %); per-cell library size correlates with mRNA content; variable genes capture biological signal rather than technical noise.

R Implementation

```r
library(Seurat)

set.seed(2026)
# Minimal simulated count matrix (300 cells x 200 genes)
counts <- matrix(rpois(300 * 200, lambda = 2), 200, 300)
rownames(counts) <- paste0("gene_", 1:200)
colnames(counts) <- paste0("cell_", 1:300)

so <- CreateSeuratObject(counts = counts, min.cells = 3, min.features = 10)
so <- NormalizeData(so, verbose = FALSE)
so <- FindVariableFeatures(so, nfeatures = 100, verbose = FALSE)
```

```r
so <- ScaleData(so, verbose = FALSE)
so <- RunPCA(so, npcs = 10, verbose = FALSE)
so <- FindNeighbors(so, dims = 1:10, verbose = FALSE)
so <- FindClusters(so, resolution = 0.5, verbose = FALSE)
so <- RunUMAP(so, dims = 1:10, verbose = FALSE)

table(Idents(so))
DimPlot(so, reduction = "umap", label = TRUE)
```

Output & Results

A `Seurat` object containing per-cell cluster assignments, dimensional-reduction coordinates (PCA, UMAP), and gene-level statistics. `DimPlot()` produces the canonical UMAP visualisation with cluster boundaries; `FeaturePlot()` overlays gene expression on the UMAP.

Interpretation

A reporting sentence: "Standard Seurat processing of the 300-cell dataset (`NormalizeData`, `FindVariableFeatures`, `ScaleData`, `RunPCA`, Louvain clustering at resolution 0.5) produced 5 clusters; cluster 3 uniquely expressed CD8A and CD8B, consistent with cytotoxic T cells, and cluster 1 expressed CD79A and MS4A1, consistent with B cells." Always describe the QC thresholds, normalisation method, number of PCs, and clustering resolution.

Practical Tips

- Run QC before normalisation; low-quality cells dominate variable-feature selection and contaminate downstream clustering.
- Use `SCTransform` instead of `NormalizeData + FindVariableFeatures + ScaleData` for a more principled variance-stabilising approach; SCTransform is the modern default for new analyses.
- The clustering `resolution` controls cluster granularity; explore values from 0.2 to 1.2 and pick based on biological interpretability rather than a single statistical criterion.
- Document Seurat version with `sessionInfo()`; the API has changed substantially between major versions (v3, v4, v5), and old code may not run on current versions.
- For very large datasets (> 100,000 cells), use Seurat v5's sketch-based subsampling or switch to the Bioconductor stack (`SingleCellExperiment`, `scran`) for memory efficiency.
- Pair clustering results with marker-gene analysis (`FindAllMarkers`) and biological annotation (SingleR, scType) before reporting cluster identities.

R Packages Used

`Seurat` for the canonical end-to-end scRNA-seq workflow; `SingleCellExperiment` for the Bioconductor data structure; `scran` and `scater` for Bioconductor-style normalisation and clustering; `harmony` for batch correction; `SingleR` for automatic cell-type annotation; `DoubletFinder` for doublet detection.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat